Proof of $BB_{SK}(7) = 41$

Boone Hapke

June 1, 2026

This note proves that the maximum normal-form size produced by any terminating SK-combinator expression of length 7 is 41. The argument combines exhaustive enumeration of all length-7 terms with a structural non-termination proof for the only remaining unresolved candidates.

1 Exhaustive Space Enumeration and Filtering

To establish the exact value of $BB_{SK}(7) = 41$, we first execute a two-phase filtering process: exhaustive state-space enumeration followed by bounded normal-order execution.

1.1 Combinator State Space Enumeration

A combinator term of length n is defined as a binary expression tree containing exactly n leaf nodes, where each leaf node is an element of the alphabet $\Sigma = \{S, K\}$. The structure of the expression is determined by the arrangement of application nodes.

The total number of unlabeled strictly binary trees with n leaves is given by the $(n-1)$-th Catalan number, C_{n-1}. For each such tree structure, every leaf node has two independent choices (S or K). Therefore, the total size of the search space for terms of length n is

$$\text{TotalTerms}(n) = 2^n C_{n-1}.$$

For $n = 7$, the sixth Catalan number is 132. The total number of valid combinator terms of length 7 is therefore

$$2^7 \cdot 132 = 16896.$$

1.2 Bounded Normal-Order Filtering

A decider program evaluates all 16896 terms using leftmost-outermost (normal-order) reduction for at most 15 steps. A reference implementation of the enumeration and filtering strategy is available in the accompanying GitHub repository placeholder: `https://github.com/USERNAME/REPO_NAME`. The terms are partitioned into two mutually exclusive categories:

- Halting terms (≤ 15 steps): terms that successfully reduce to a normal form (a state containing no further reducible expressions, or redexes) within 15 steps. The maximal combinator-count normal form observed among all terminating terms in this space is 41 combinators, achieved by the normal form of the term $SSS(S(SS))S$.
- Unresolved terms: terms that have not reduced to a normal form after 15 reduction steps.

Out of the 16896 total candidates, exactly two remain unresolved after 15 steps of reduction:

$$T_1 = S(SS)SSSS,$$
$$T_2 = SSS(SS)SS.$$

1.3 Equivalence via a Single Reduction Step

Notational note: throughout this proof, the reduction arrow $\xrightarrow{n}$ and step count n refer strictly to standard normal-order (leftmost-outermost) reduction, unless otherwise specified.

It is immediate that

$$T_1 \xrightarrow{1} SSS(SS)SS = T_2.$$

Because T_1 reduces to T_2 in exactly one normal-order step, their termination behaviors are identical:

$$T_1 \text{ is non-terminating} \iff T_2 \text{ is non-terminating.}$$

Consequently, proving the termination behavior of $T_1 = S(SS)SSSS$ simultaneously settles the outcome for both unresolved candidates.

1.4 Proof Strategy

To prove non-termination of $T_1 = S(SS)SSSS$, we construct an infinite family of expressions M_n such that each instance reduces into a larger structure containing the next instance. By establishing that the initial term $S(SS)SSSS$ can be reduced to an expression of the form FM_0 (where F and M_0 are explicitly defined combinator terms), and that each M_n generates an infinite reduction chain, we demonstrate that $S(SS)SSSS$ itself admits an infinite reduction path, proving non-termination.

2 Proof of Non-Termination for $S(SS)SSSS$

2.1 Global Lemmas

Lemma 1 (Substitution Principle)**.** *If a combinator expression A reduces to an expression B in any number of steps, $A \xrightarrow{*} B$, then for any structural tree* *context* $\mathcal{C}[\cdot]$,

$$\mathcal{C}[A] \xrightarrow{*} \mathcal{C}[B].$$

Proof. Primitive reduction rules apply strictly to local subtrees that match a designated structural pattern. Because the evaluation of a local redex depends only on its internal tree structure and not on its parent or sibling branches, embedding the tree inside a larger application context $\mathcal{C}[\cdot]$ preserves the ability to perform all internal reductions. □

Lemma 2. *In the pure S-calculus, non-terminating expressions cannot be erased. If an expression contains a non-terminating subterm, then the entire expression is non-terminating under every reduction strategy.*

Proof. The S-rule

$$Sxyz \to xz(yz)$$

retains every input component x, y, and z on its right-hand side. Because the pure S-calculus lacks erasing combinators such as K, any non-terminating subterm persists through reduction and prevents the parent expression from reaching a normal form. □

2.2 Preliminary Definitions

Let x be a fixed dummy variable. Define

$$\begin{aligned} F &= S(S(SS)S)S, \\ D &= S(S(S(S(SS)F))), \\ M_0 &= Dx, \\ M_n &= SM_{n-1}\big(x(SM_{n-1})\big) \quad \text{for } n \geq 1. \end{aligned}$$

2.3 Inductive Cycle

We establish that the family of expressions M_n forms an infinite reduction cycle by breaking the transformation down into three phases.

Phase A: Expansion of FM_n. For any n, the application of FM_n unfolds to reveal a self-application pattern:

$$FM_n \xrightarrow{*} \mathcal{C}_0\big[M_n(SM_n)\big].$$

By expanding the definition of F and applying it to an arbitrary term t, a direct three-step normal-order reduction gives

$$Ft \xrightarrow{3} \mathcal{C}_0\big[t(St)\big].$$

Substituting M_n for t produces the claimed context.

Phase B: Reduction to the base case. Any expression M_n applied to SM_n reduces through successive applications of M_n to the base case:

$$M_n(SM_n) \xrightarrow{*} \mathcal{C}_1\big[M_0(SM_n)\big].$$

By definition,

$$M_n = SM_{n-1}\big(x(SM_{n-1})\big).$$

Applying this definition to an arbitrary term t yields

$$M_n t \to M_{n-1} t\big(x(SM_{n-1})t\big).$$

Every application of $M_n t$ thus structurally reduces to an application of $M_{n-1}t$, and induction on n shows the expression unrolls until it reaches the base term M_0. Substituting SM_n for t gives the claimed target context.

Phase C: Index advancement. The base term M_0 increments the family index when it is applied to SM_n:

$$M_0(SM_n) \xrightarrow{*} \mathcal{C}_2[FM_{n+1}].$$

Using $M_0 t = Dxt$, after four reduction steps the base application produces the context $F(t(xt))$:

$$Dxt \xrightarrow{4} \mathcal{C}_2[F(t(xt))].$$

Substituting SM_n for t gives

$$M_0(SM_n) \xrightarrow{*} \mathcal{C}_2\big[F\big(SM_n(x(SM_n))\big)\big].$$

By the definition of the inductive family, the subterm $SM_n(x(SM_n))$ is exactly M_{n+1}, so the reduction simplifies to

$$M_0(SM_n) \xrightarrow{4} \mathcal{C}_2[FM_{n+1}].$$

Conclusion of the inductive cycle. Combining Phases A–C yields the composed reduction

$$FM_n \xrightarrow{*} \mathcal{C}_0\big[M_n(SM_n)\big] \xrightarrow{*} \mathcal{C}_1\big[M_0(SM_n)\big] \xrightarrow{*} \mathcal{C}_2\big[FM_{n+1}\big],$$

so that one full cycle transforms FM_n into a context containing FM_{n+1}. Therefore, by induction on n the family produces an infinite sequence of index increments

$$\mathcal{C}_a[FM_0] \xrightarrow{*} \mathcal{C}_b[FM_1] \xrightarrow{*} \mathcal{C}_c[FM_2] \xrightarrow{*} \cdots,$$

showing that the subterm FM_0 admits an infinite reduction path.

2.4 Initialization and Main Theorem

Theorem 1. *The pure S-combinator term $S(SS)SSSS$ is non-terminating.*

Proof. Begin by executing the first nine normal-order reduction steps of the initial term:

$$S(SS)SSSS \xrightarrow{9} S(S(SS)F)\big(F(S(SS)F)\big).$$

Write the left-hand subterm as $\mathcal{L} = S(S(SS)F)$ and the right-hand subterm as $\mathcal{R} = F(S(SS)F)$. For an S combinator to form a redex, it must have at least three arguments. In the global expression $\mathcal{L}\mathcal{R}$, the head combinator $\mathcal{L}$ has only one argument, so top-level application cannot reduce. Furthermore, the internal subterm $S(SS)F$ has only two arguments and is therefore inert.

Because the left branch $\mathcal{L}$ is in normal form and cannot be reduced, normal-order reduction is forced to evaluate only the right-hand subterm. Empirical tracking of $\mathcal{R}$ shows that after exactly 62 evaluation steps, the expression expands to

$$F(S(SS)F) \xrightarrow{62} \mathcal{C}_a[F(DD)].$$

The subterm DD represents the base term $M_0 = Dx$ with the dummy variable x instantiated as D. Since the behavior of the inductive cycle does not depend on the specific choice of x, we may write this term as M_0, yielding

$$F(S(SS)F) \xrightarrow{62} \mathcal{C}_a[FM_0].$$

We have proven inductively that the subterm FM_0 possesses an infinite reduction path. By Lemma 2, because the pure S-calculus cannot erase subexpressions, the existence of this infinite reduction path within $\mathcal{C}_a[FM_0]$ guarantees that the parent expression $\mathcal{R}$ is non-terminating under any reduction strategy. Consequently, the initial term $S(SS)SSSS$ is non-terminating. □

3 Conclusion

By coupling the exhaustive filtering with the structural induction proof showing that T_1 unrolls into an infinitely growing cycle

$$\mathcal{C}_a[FM_0] \xrightarrow{*} \mathcal{C}_b[FM_1] \xrightarrow{*} \mathcal{C}_c[FM_2] \xrightarrow{*} \cdots,$$

we prove that the only two unresolved terms in the length-7 term space are non-terminating. Because all other candidate terms of length 7 have been verified to halt in ≤ 15 steps, the term $SSS(S(SS))S$, which reduces to a term of length 41, is definitively the term with the maximum normal form size. Given that $BB_{SK}(n)$ is defined as the maximum term length of a normal form generated by a halting term of length n, and the normal form of $SSS(S(SS))S$ contains exactly 41 combinators, the seventh BB_{SK} value is established:

$$BB_{SK}(7) = 41.$$